

M2 TIW – M2 BIO-INFO

DATA ANALYSIS

Other Data Types Transformations

DATA TRANSFORMATION

- Our data is provided in a given form
 - Tabular (vectors)
 - Network
 - Time series
 - Text
 - Images
 -
- To use the full potential of data mining, you might want to study it from multiple angles
 - How to convert from tabular to graph?
 - From Graph to Tabular?
 - From images/text to tabular (embedding)?

DIMENSIONALITY REDUCTION

Low dimensionality embedding

DIMENSIONALITY REDUCTION

- Data Mining objective: understand our data
 - We get a dataset composed of many features
 - Or worst, complex object (image, sound, graph...)
 - How to understand the organization of our data?
 - How to perform clustering?

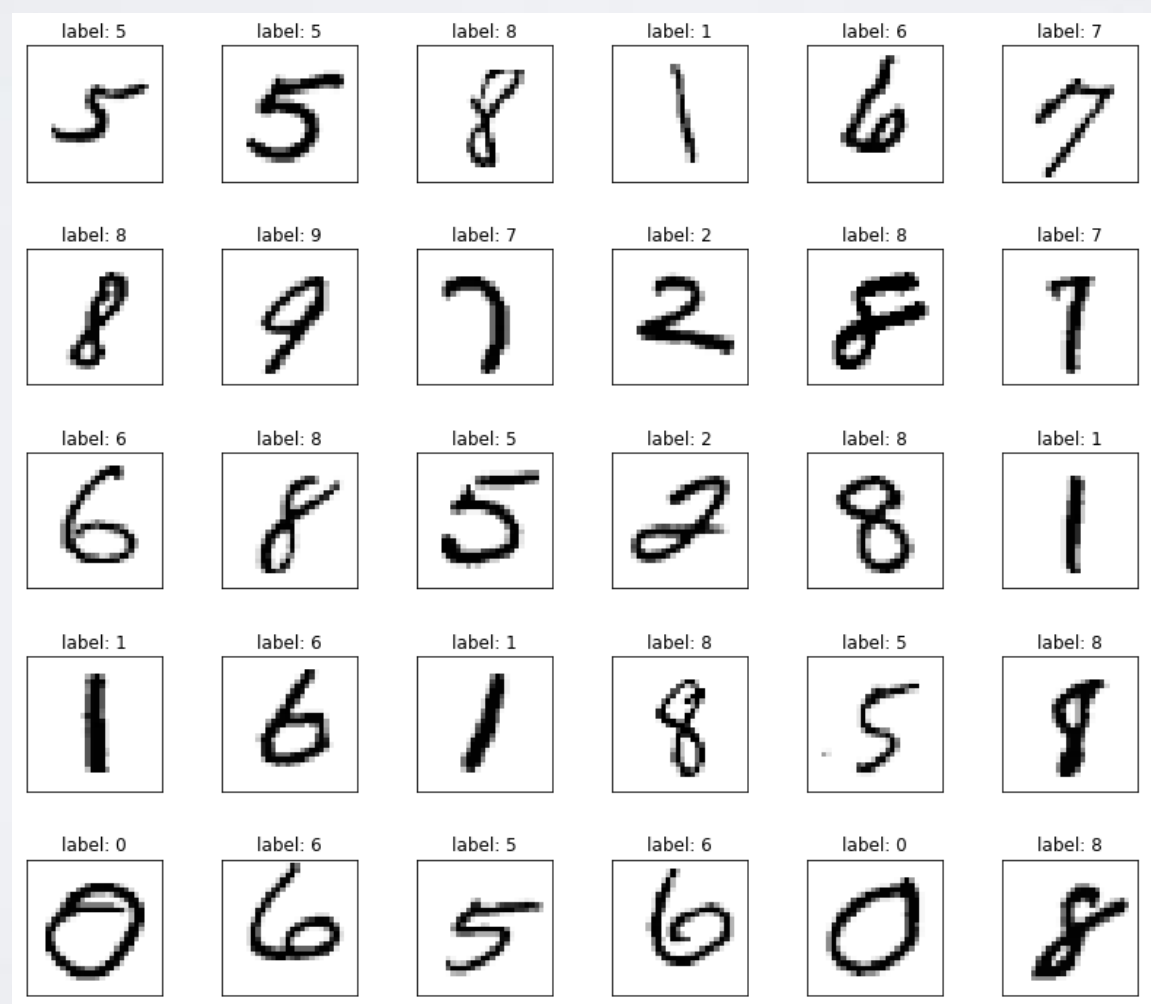
VISUALIZATION

- Your data is perfectly fine, but you want to intuitively understand how it is organized
 - Are there groups of similar objects?
 - Are my clusters meaningful?
 - Is my classification/clustering on some types of elements and not others.

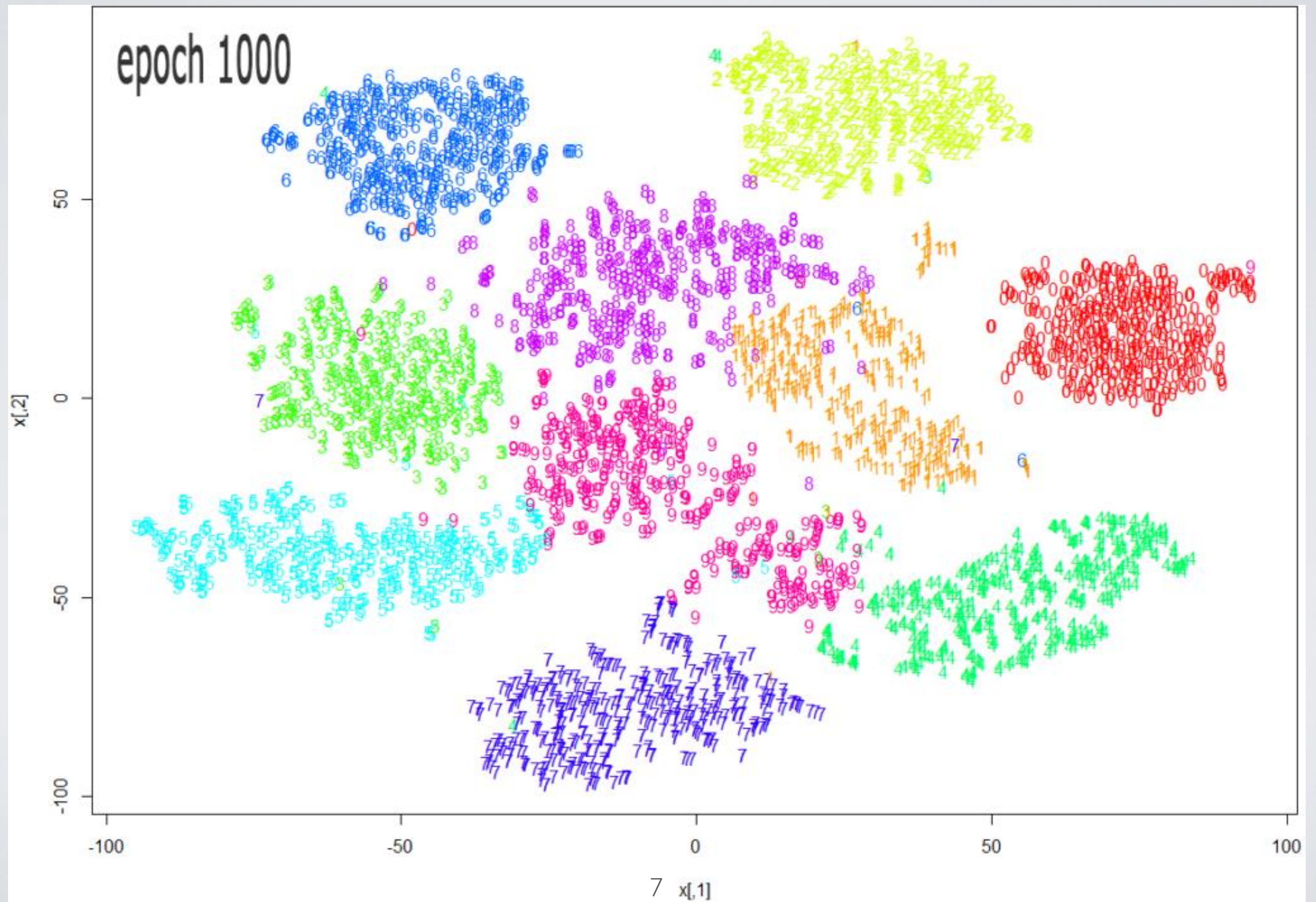
VISUALIZATION

Example: MNIST Dataset

Each pixel is a variable



t-SNE embedding



CURSE OF DIMENSIONALITY

- Having hundreds/thousands of attributes is a problem for data analysis.
 - e.g.: medicine: blood analysis, genomics....
 - e.g.: cooking recipes: each column an ingredient...
- We want to reduce the number of attributes while keeping most of the information
- Also helps with scalability

CORRELATION

- Assume that you have correlated features such as age, height and weight.
 - Redundancy ! Computational Inefficiency
 - e.g., Decision tree will spend a lot of time choosing between them for no reason
 - Risk of overfitting
 - noise between correlated variables used to distinguish individuals
 - Model interpretability
 - e.g., a model will say that y depends on x or w randomly, if x and w correlated
- Dimensionality reduction can create a single variable to capture what is common
 - The rest can be lost or captured by another feature,
 - Engine horsepower, Car weight, Fuel Consumption
 - =>Performance index (horsepower and weight)
 - =>Efficiency score (weight and fuel consumption)

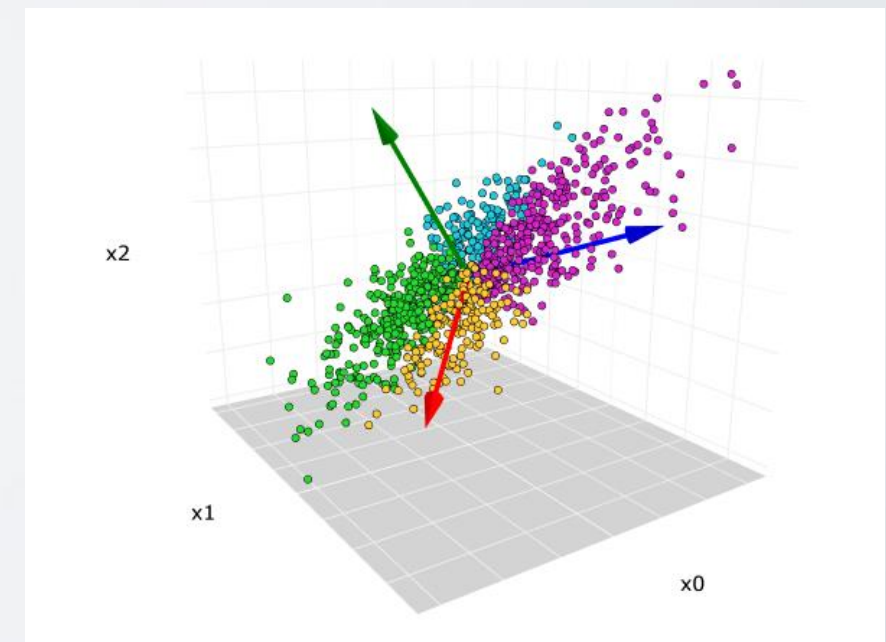
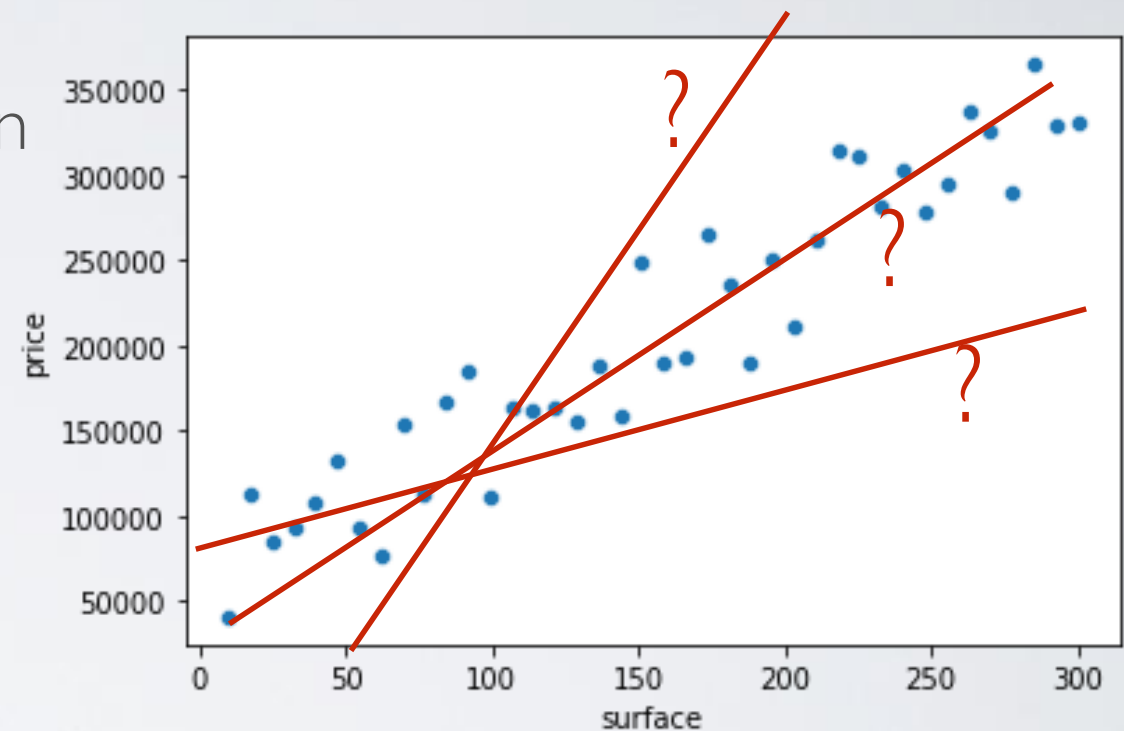
PCA

PCA

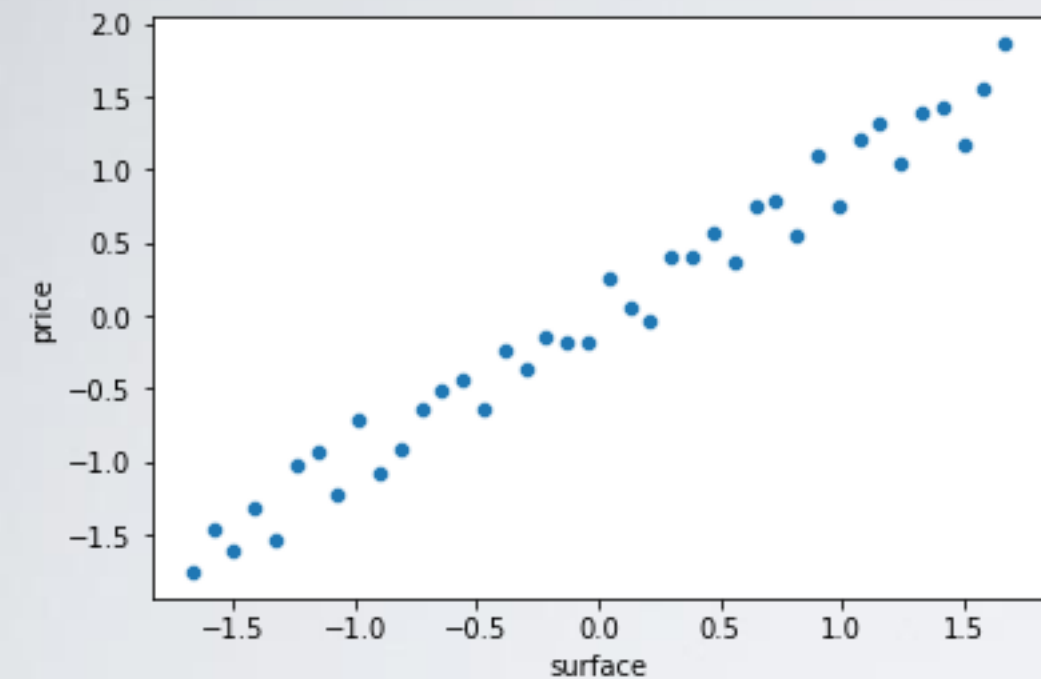
- PCA: Principal Component Analysis
- Defines new dimensions that are linear combinations of initial dimensions
 - Objective: concentrate the **variance** on some dimensions
 - So that we can keep only these ones.
 - Those we remove contain low variance, thus low information

PCA

- Algorithm:
 - 1) Find an “axis”, a unit vector defining a line in the space
 - That minimizes the variance $\Rightarrow$ the squared distance from all points to that line
- 2) **For** d **in** $[2:(\text{initial_d})]$
 - Find another axis, with two constraints:
 - Orthogonal to all previous axis
 - Among those, minimizing the variance
- 3) At the end, keep the first k dimensions
 - Some information is lost



EXAMPLE PCA 2D

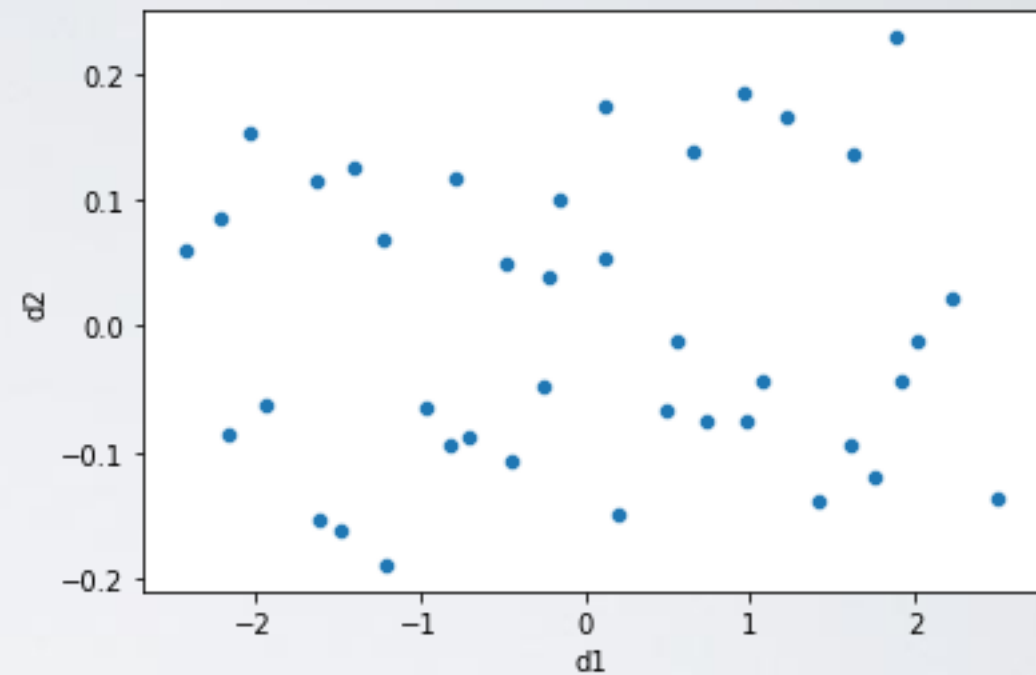


Covariance matrix (original)

```
[1.      , 0.98675899],
 [0.98675899, 1.      ]
```

Sum of variance
2

Variance by dimension
1 1



Covariance matrix (pca)

```
[ 1.98675899e+00, 0],
 [0, 1.32410092e-02]
```

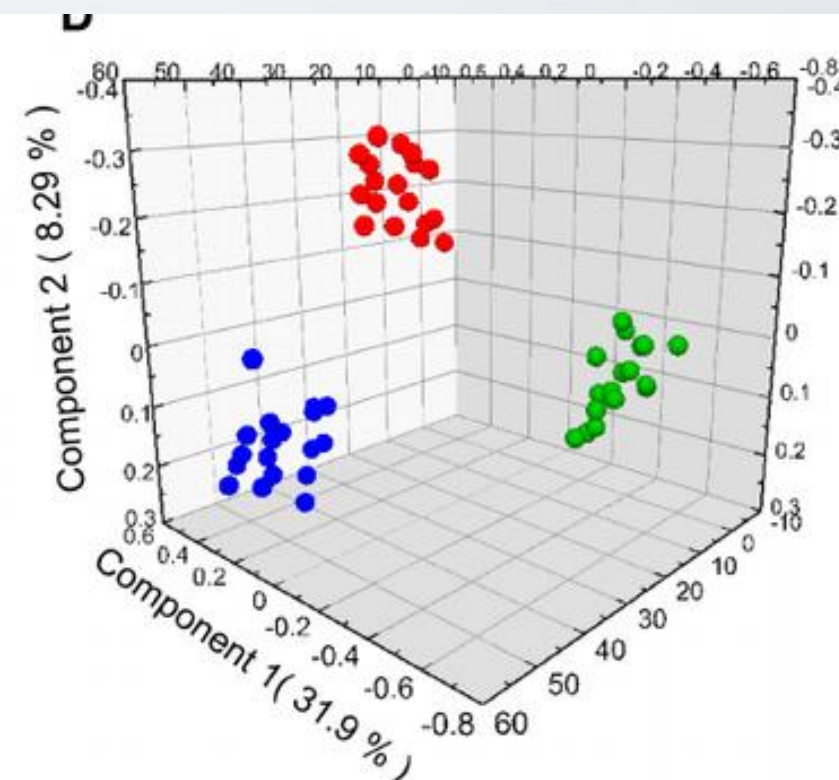
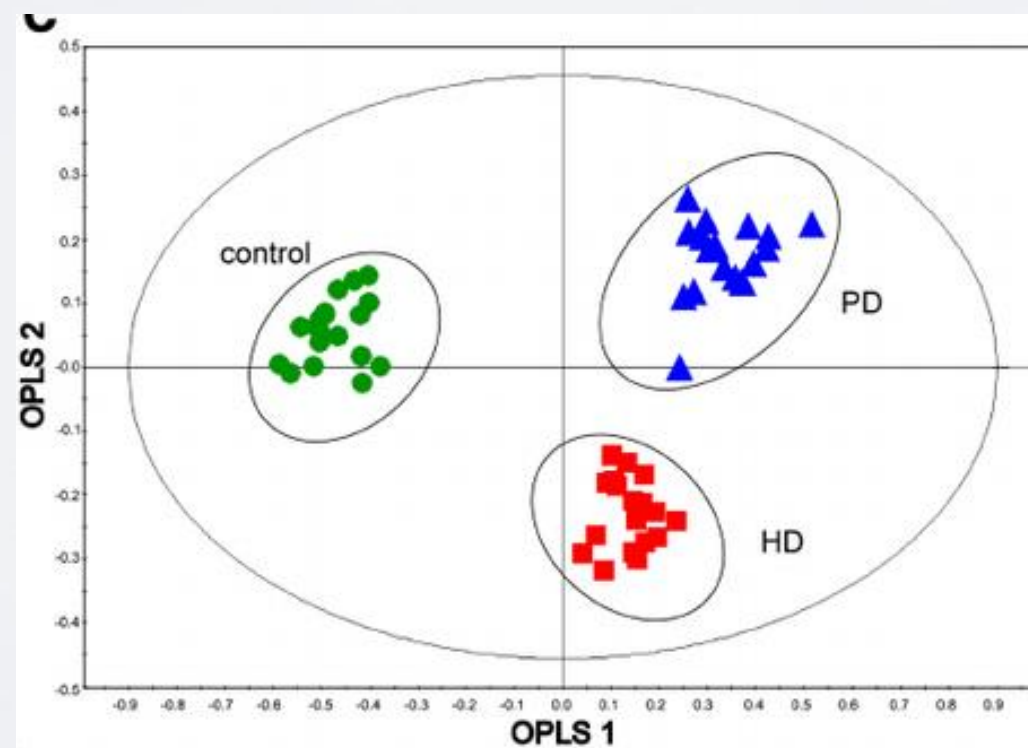
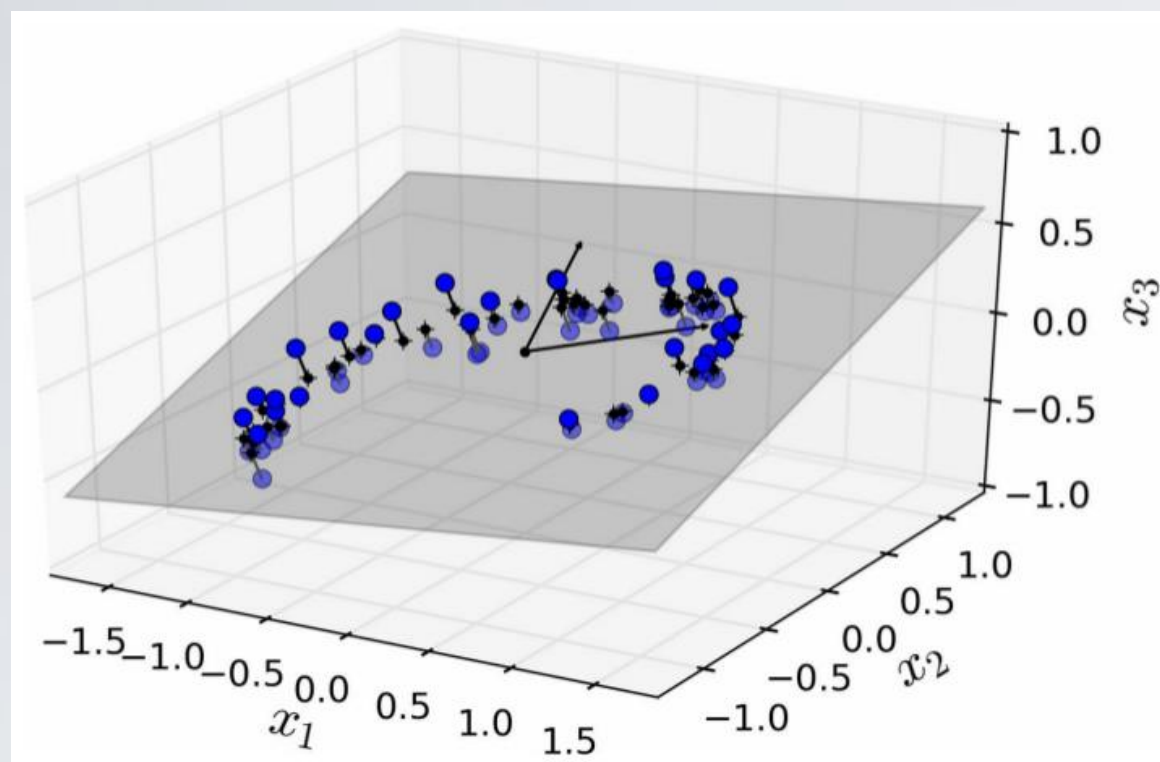
Sum of variance
2

Variance by dimension
1.98675899 0.01324101

Explained variance(ratio)
13

```
[0.9933795, 0.0066205]
```

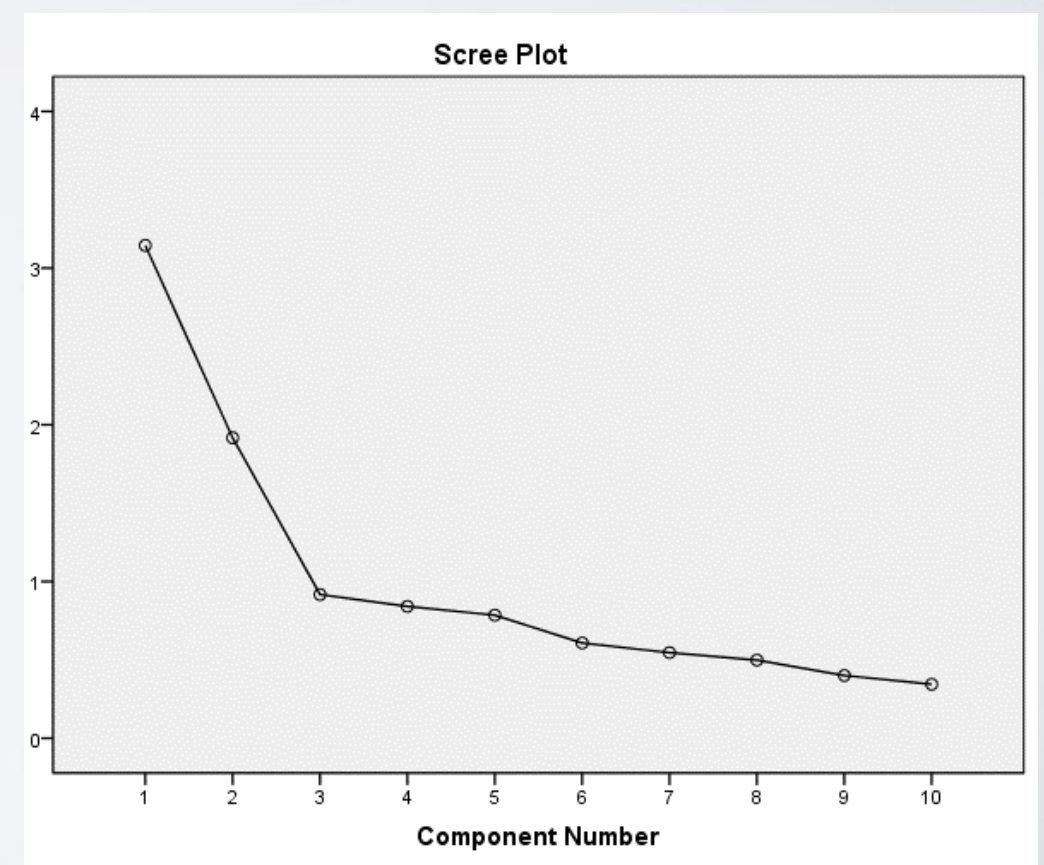
3D=>2D



CHOOSING COMPONENTS

- How to choose k?
 - Elbow method... BIC/AIC...
 - OR fix beforehand a min threshold of explained variance, e.g.: 80%
 - We are fine with losing 20% of information
 - If there is a downstream task, cross-validation

Explained
variance



COMPUTATION IN PRACTICE

- From standardized dataset X
- Method 1:
 - 1) Compute the Covariance Matrix ($X^T X$)
 - => Linear Correlation Matrix
 - 2) Find the eigenvectors of this matrix
 - $X^T X = V \Lambda V^T$
 - V : eigenvectors = Principal components, Λ : Eigenvalues, = explained variance
- Method 2:
 - Apply SVD matrix decomposition
 - $X = U \Sigma V^T$
 - U : left singular vectors. Σ : diagonal matrix with the singular values, V^T : right singular vectors (the principal components)

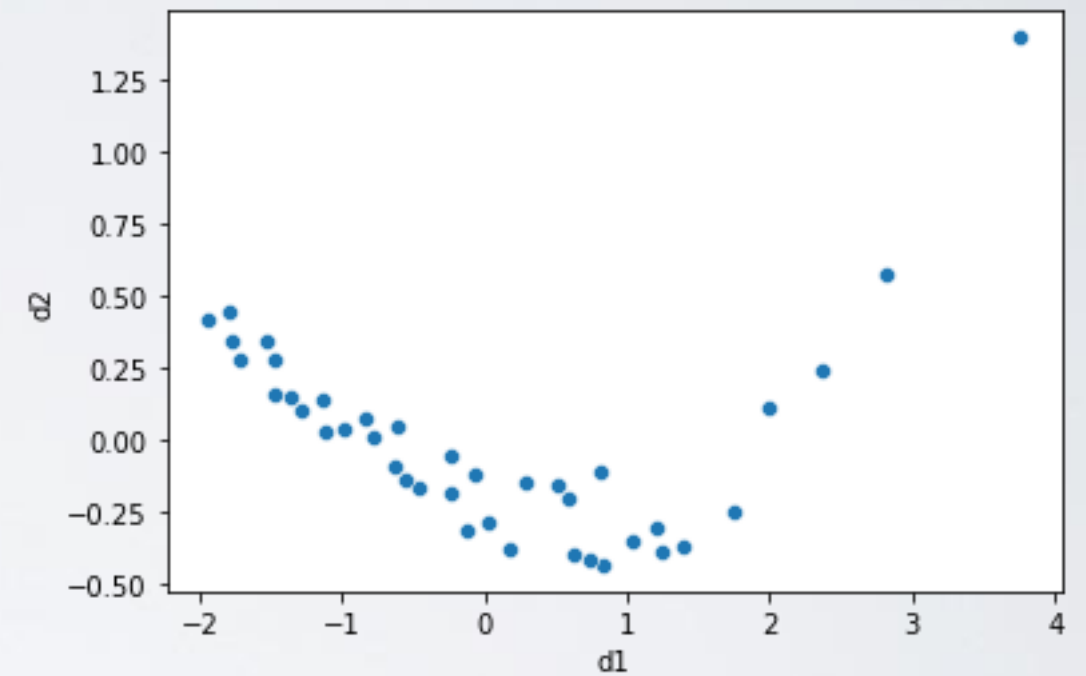
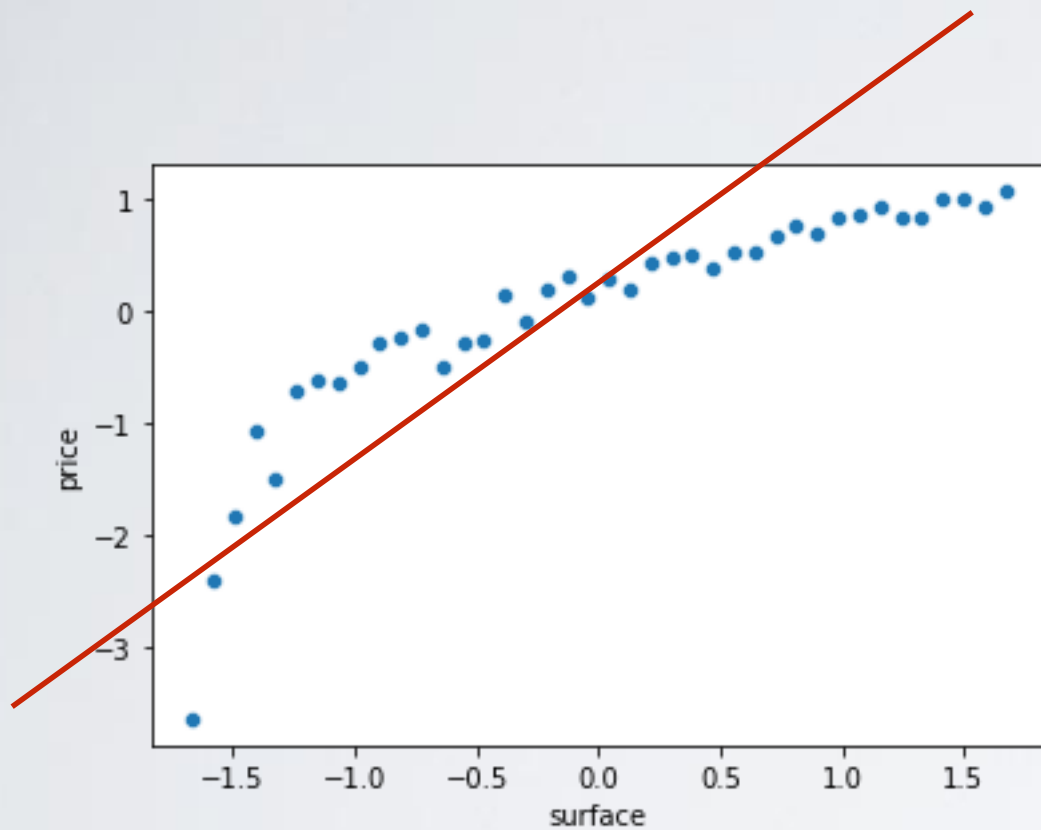
COMPUTATION IN PRACTICE

- V are the principal components
- Computing the new positions for each observation:
 - XV

PCA POPULARITY

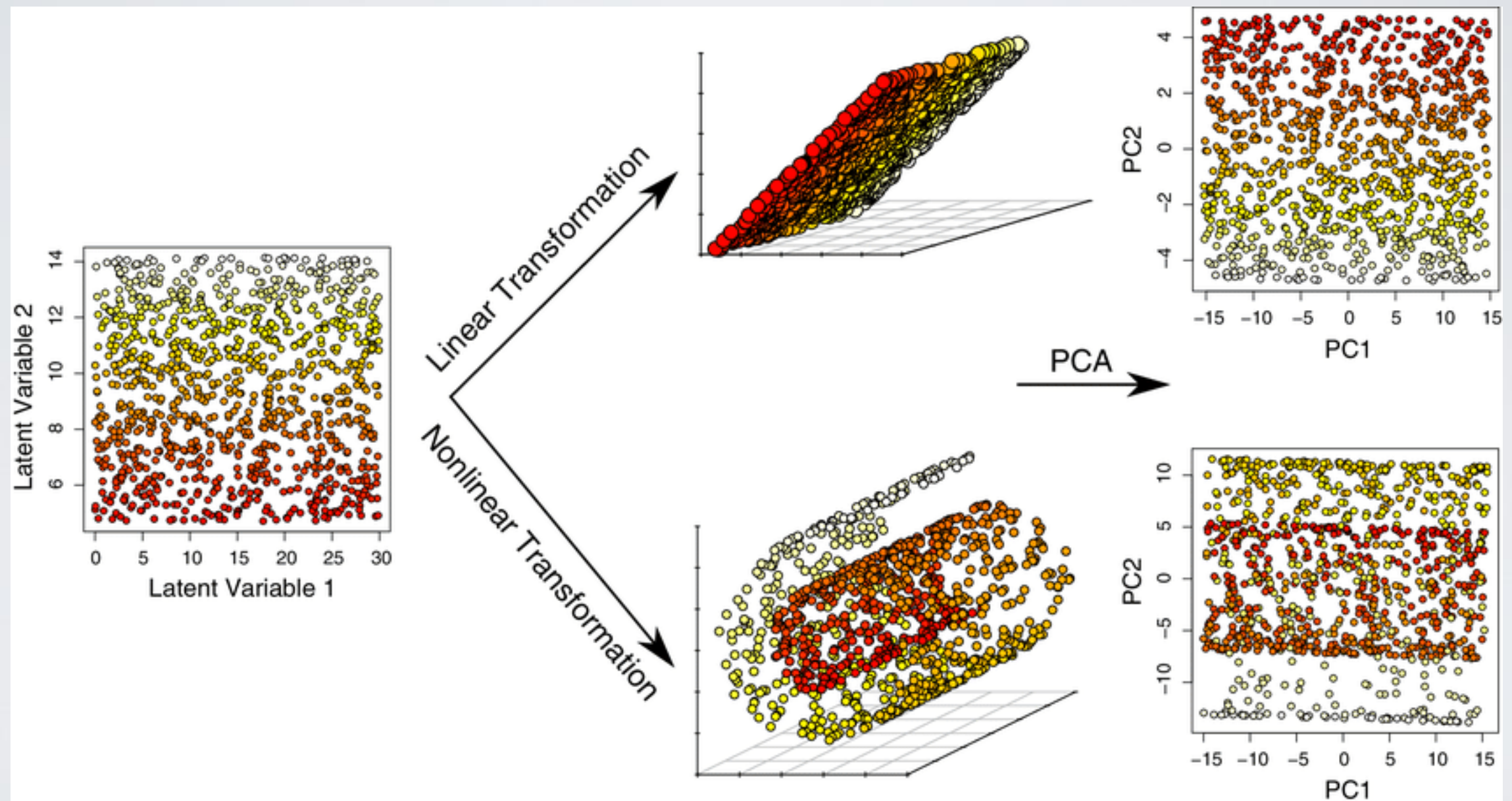
- Why is PCA popular?
- Similar reasons than linear regression:
 - Useful
 - Eliminate correlations
 - Analytical solutions
 - Guarantee to find the global minimum of the objective
 - Could be done before modern computers
 - Interpretable solution
 - Intuitively pleasant
- No reason to consider it “better” than other methods for dimensionality reduction...

NON-LINEAR SITUATIONS



Pearson correlation(d1,d2): 0

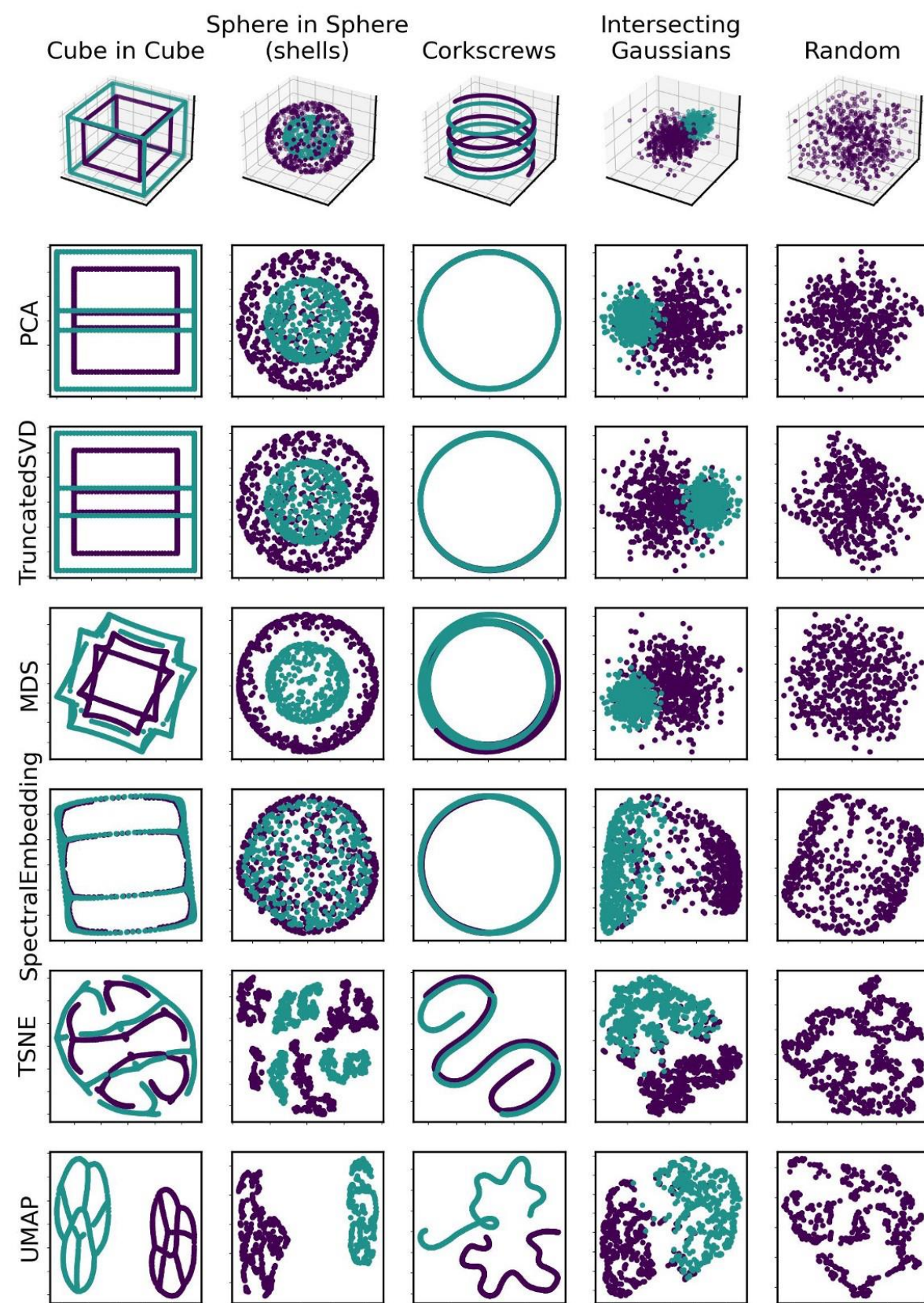
NONLINEAR DATA



MANIFOLDS

MANIFOLDS

- Manifolds are another approach to dimensionality reduction
- The general principle is to
 - 1) Define a notion of distance between elements in the original space
 - 2) Define a notion of distance between elements in a reduced, target space
 - 3) Minimize the difference between distances in original and target space
- In many cases, the process is nonlinear, i.e., we choose distances such as
 - We care more about preserving the distance for items “close” in space than for those “far” from each other

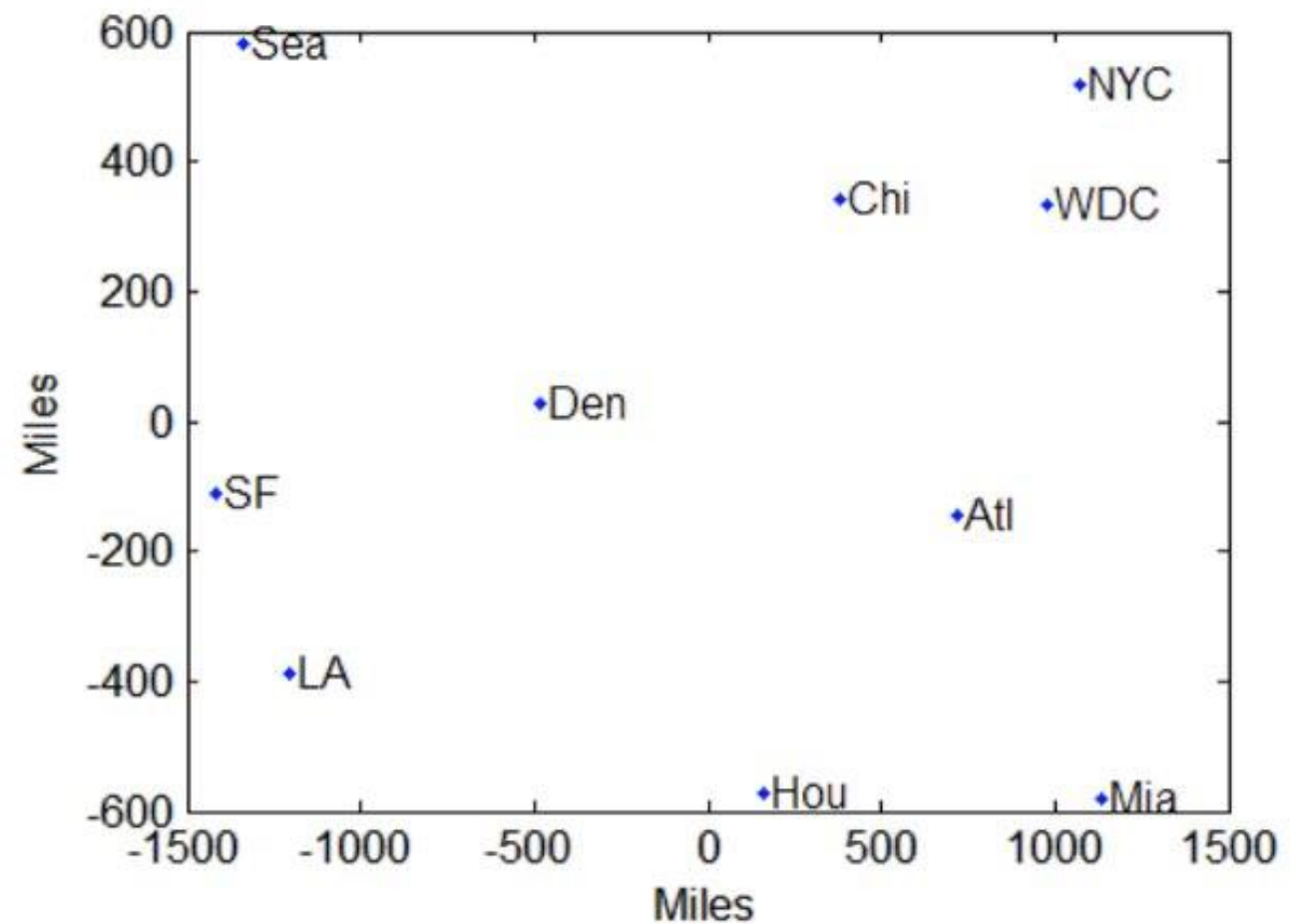


MDS

- MDS: Multi-dimensional Scaling:
 - Simply minimize distance between original space and target space
 - e.g., d-dimensional forced to 2-dimensional
- How to do it?
 - 1) Compute all (squared Euclidean) pairwise distances between items=> **Similarity matrix**
 - $n \times f$ matrix => $n \times n$ matrix
 - Apply double-centering (remove row and column means)
 - 2) Compute PCA on this similarity matrix
- Problems:
 - Very costly (nb features=nb elements), n^2
 - Try to preserve all distances, therefore extremely constrained

MDS

	Atl	Chi	Den	Hou	LA	Mia	NYC	SF	Sea	WDC
Atl	0	587	1212	701	1936	604	748	2139	2182	543
Chi	587	0	920	940	1745	1188	713	1858	1737	597
Den	1212	920	0	879	831	1726	1631	949	1021	1494
Hou	701	940	879	0	1374	968	1420	1645	1891	1220
LA	1936	1745	831	1374	0	2339	2451	347	959	2300
Mia	604	1188	1726	968	2339	0	1092	2594	2734	923
NYC	748	713	1631	1420	2451	1092	0	2571	2408	205
SF	2139	1858	949	1645	347	2594	2571	0	678	
Sea	2182	1737	1021	1891	959	2734	2408	678	0	
WDC	543	597	1494	1220	2300	923	205	2442	2329	

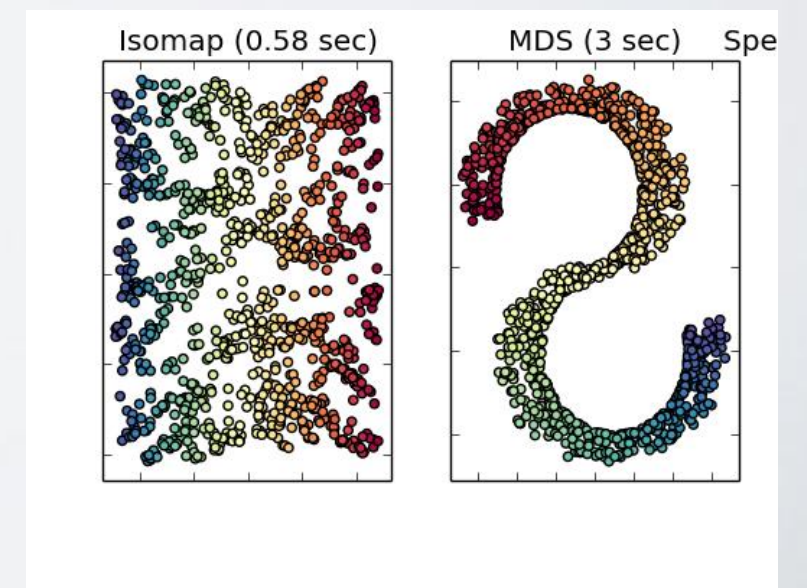
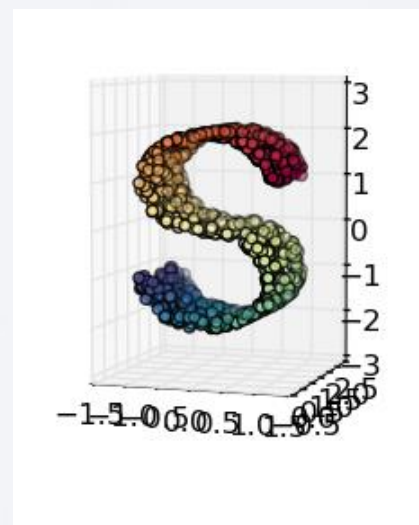


ISOMAP

- Variation of MDS

- ▶ 1) We define a **graph** such as two elements are connected if they are at $\text{distance} < \text{threshold}$. (Alternative: fixed number of neighbors)
 - Put a weight on edges = euclidean distance
- ▶ 2) Compute a **similarity** matrix, such as $\text{distance} = \text{weighted shortest path distance}$
- ▶ 3) Apply MDS on it

- Non-linear distances



T-SNE

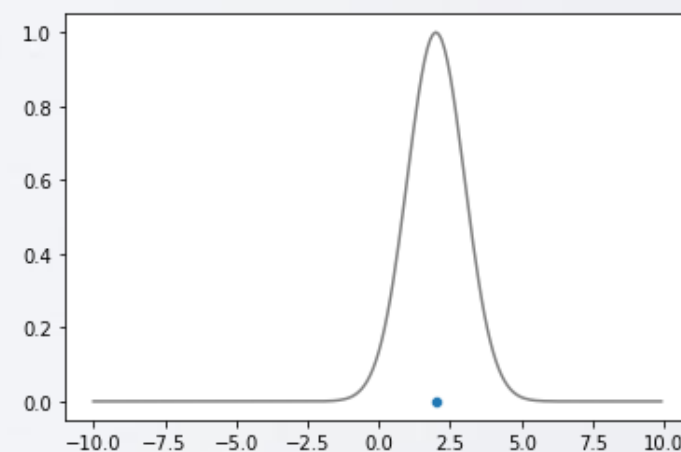
T-SNE

- t-SNE : t-distributed stochastic neighbor embedding
- Non-linear dimensionality reduction
- One of the most popular method for visualizing data in low dimensions
- Similar to MDS/Isomap, but:
 - Do not try to preserve long distance at all
 - Can preserve local structure but loose global one
 - Optimized via gradient descent
 - No way to guarantee global optimum

SNE

- General principle:

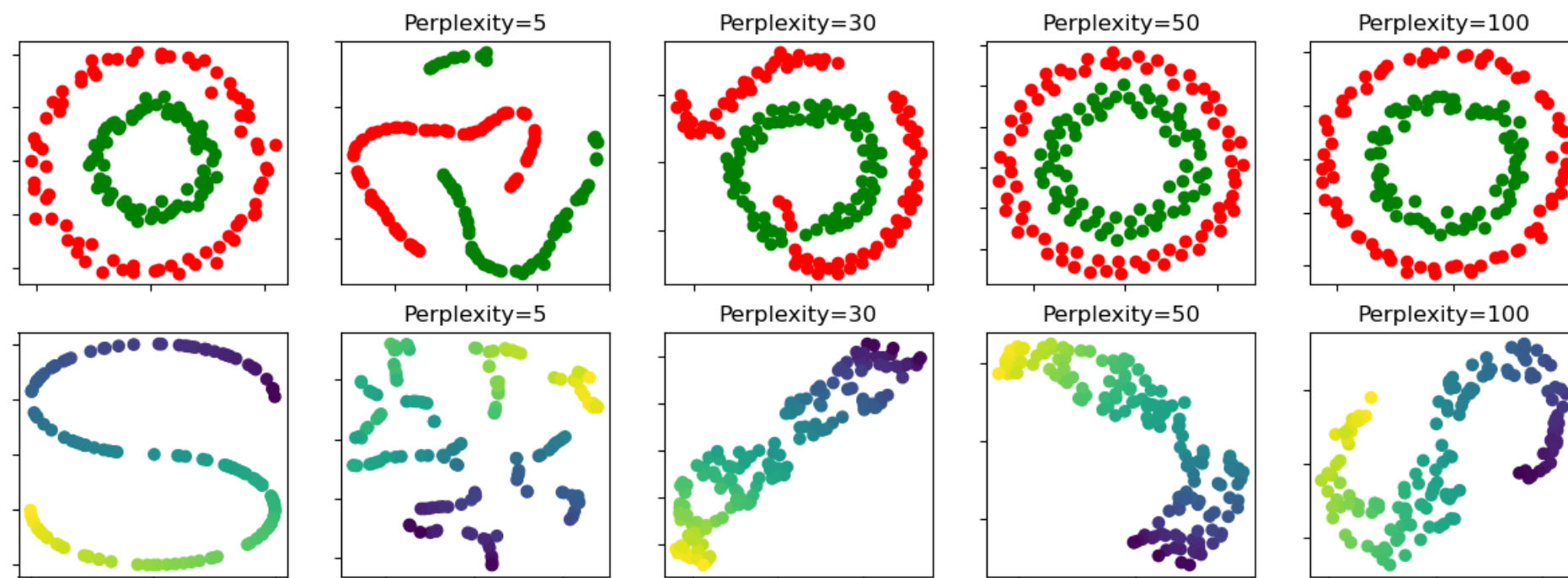
- ▶ Define a notion of similarity $p_{j|i}$ in the high dimensional space P
 - Based on normal distribution
- ▶ Define a notion of similarity $q_{j|i}$ in the low dimensional space Q
 - Based on student-t distribution, tends to “exaggerate” differences
- ▶ For each point of initial coordinates $\mathbf{x}_i$, find a new coordinate $\mathbf{y}_i$ in the lower dimensional space, such as to minimize the difference between P and Q
 - $\forall i,j p_{j|i} \approx q_{j|i}$



T-SNE: PERPLEXITY

- There is a perplexity parameter σ : it controls how much each point cares more about close neighbors compared with farther neighbors
 - Low σ : Preserve mostly local distances
 - High σ : Give more importance to long-range distances
 - More expensive, more similar to MDS

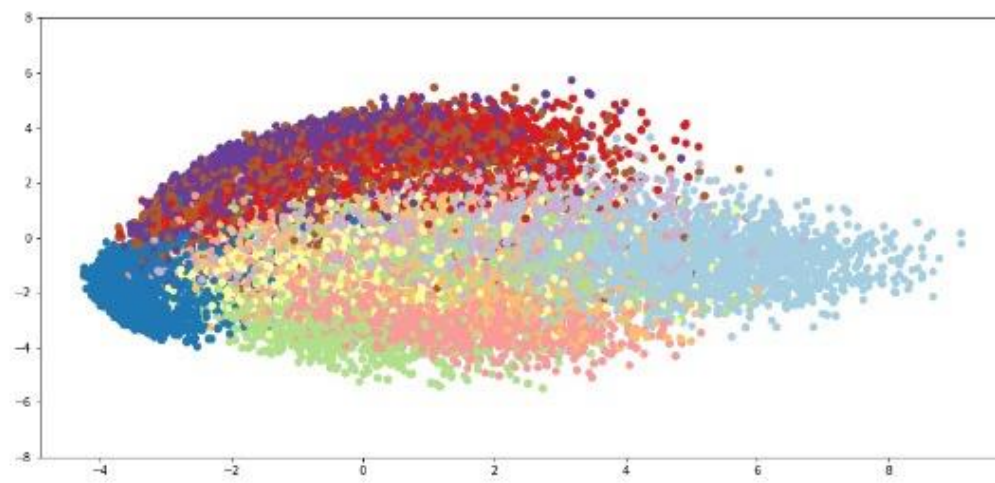
INFLUENCE OF PERPLEXITY



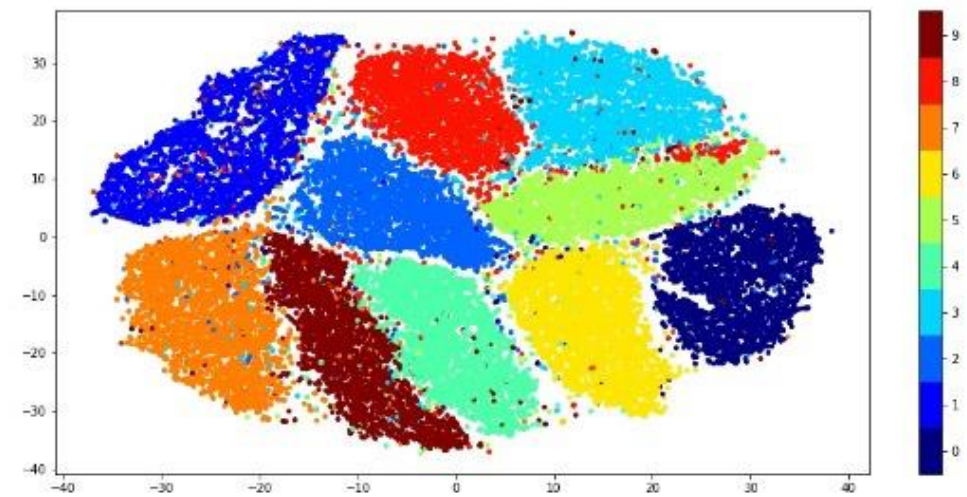
More like
Isomap

More like
MDS

MNIST - PCA



MNIST - TSNE



LOW DIMENSIONAL EMBEDDINGS

EMBEDDINGS

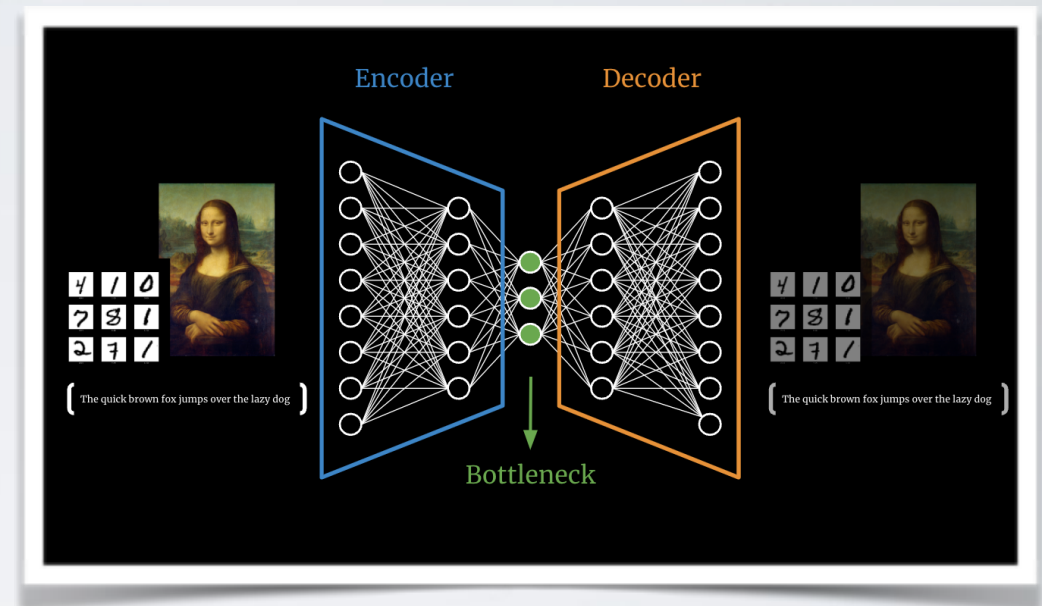
- A recent usage of low dimensional embeddings is to encode complex objects as vectors
 - Words as Vector $\Rightarrow$ Word2Vec
 - Nodes (of graph) as Vectors $\Rightarrow$ Node2Vec
 - Documents as Vectors $\Rightarrow$ Doc2Vec
 -

NODE2VEC

- Shallow neural network method
- Objective similar to MDS/tSNE:
 - Minimize the difference between **graph distance** and **embedding distance**
 - Parameter to tune local/long-distance graph distance
 - (Example implementation: <https://github.com/eliorc/node2vec>)

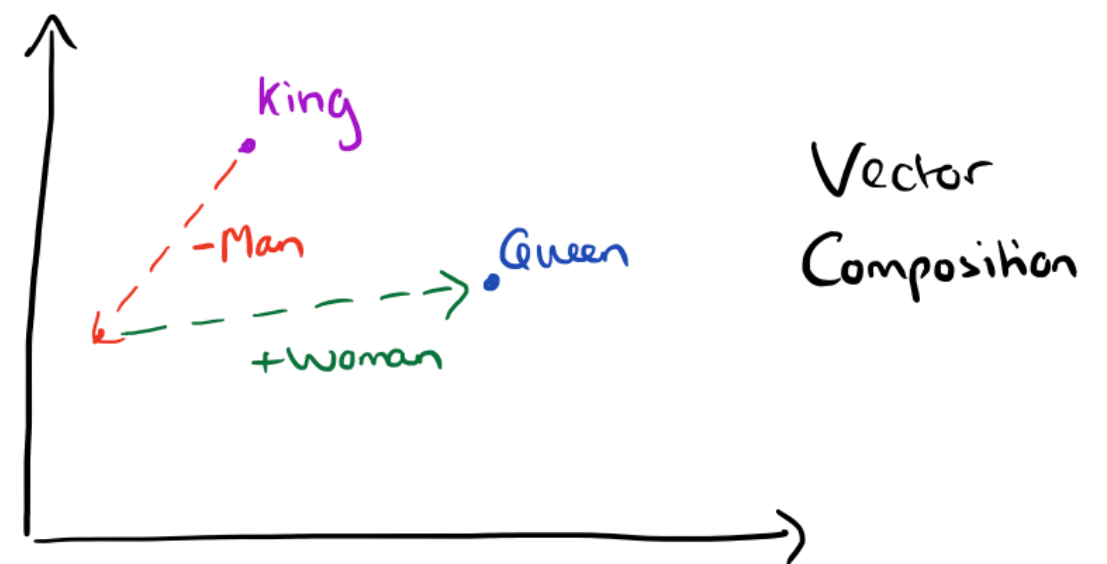
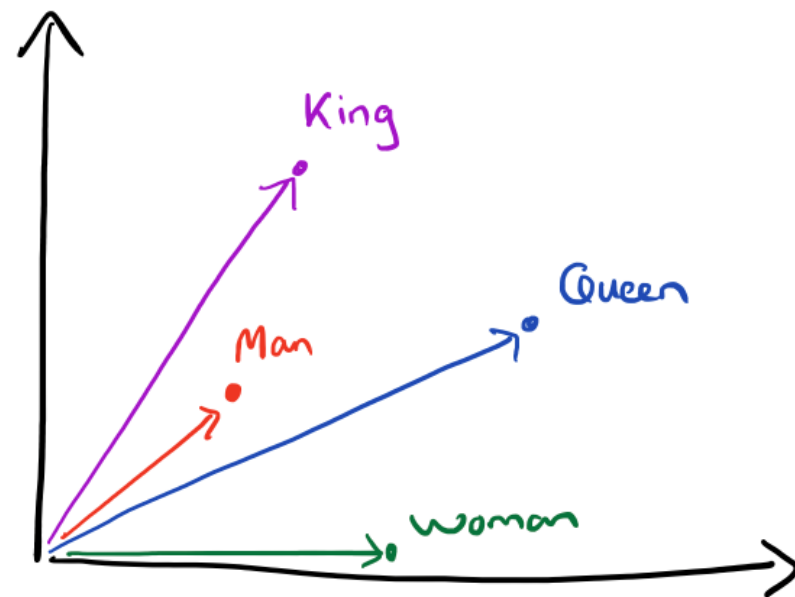
AUTO-ENCODER

- Deep learning approach
- Autoencoder:
 - DECODER neural network learns to reconstruct an input object from a small vector
 - ENCODER neural network learns to encode an input object into a small vector (to maximize reconstructability)
- Created for images, work similarly for texts or graphs
 - (Variational GAE)



EMBEDDINGS

	King	Queen	Woman	Princess	...
Royalty	0.99	0.99	0.02	0.98	
Masculinity	0.99	0.05	0.01	0.02	
Femininity	0.05	0.93	0.999	0.94	
Age	0.7	0.6	0.5	0.1	
...	...				



<https://blog.acolyer.org/2016/04/21/the-amazing-power-of-word-vectors/>

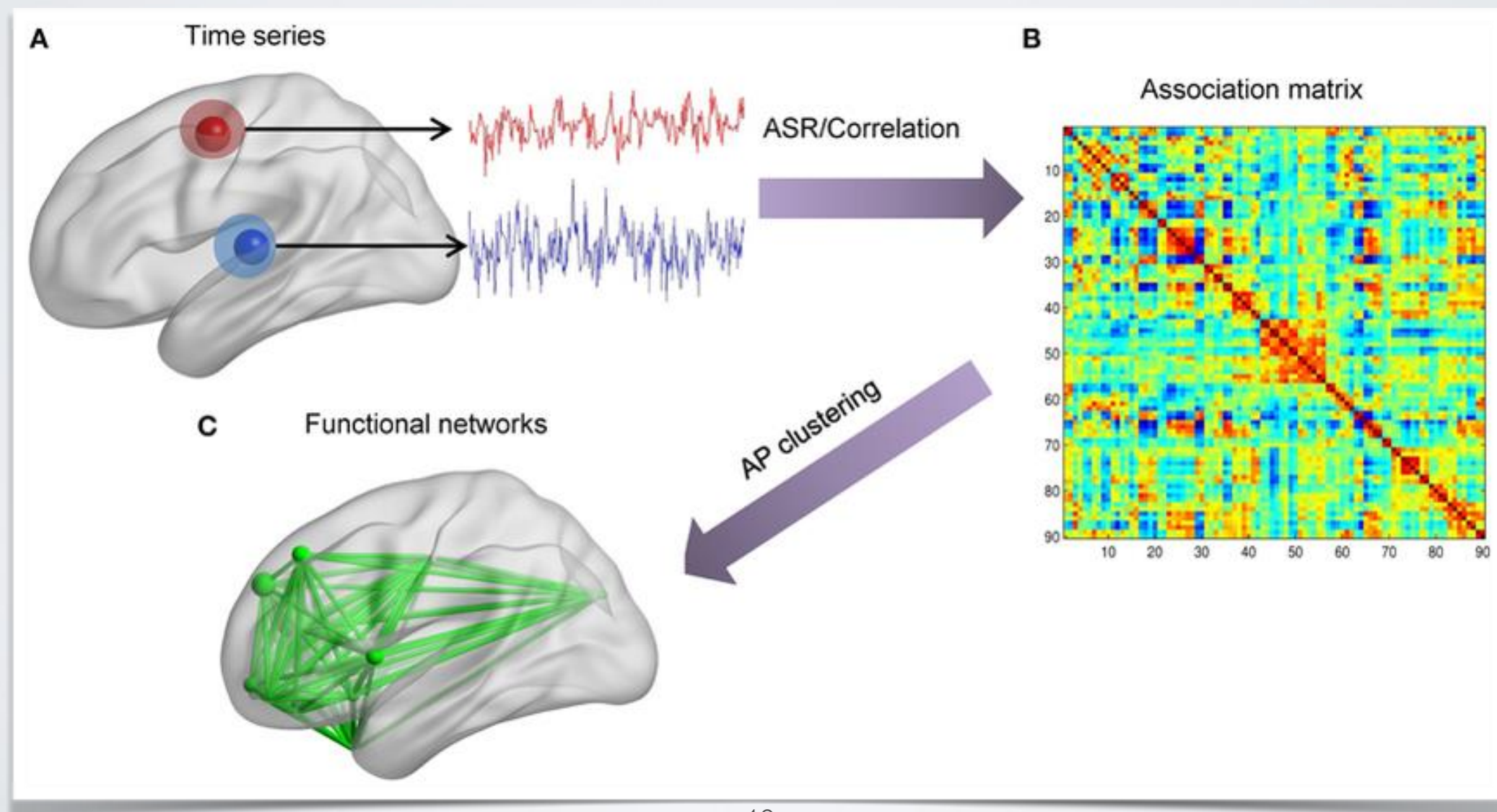
OBJECTS/VECTORS TO GRAPHS

GRAPH $\leftrightarrow$ VECTORS

- Graph Embedding: Graph $\rightarrow$ Vectors
- What about Vectors $\rightarrow$ Graphs
 - Simple approach: Correlation matrix
 - $\Rightarrow$ Represent the relations between features in a dataset
 - 1) Compute the correlation between all variables (spearman/Pearson)
 - 2) Keep only correlations above a threshold (alternative: x% strongest)
 - 3) Correlation values can be represented as weights

ITEM-ITEM GRAPH

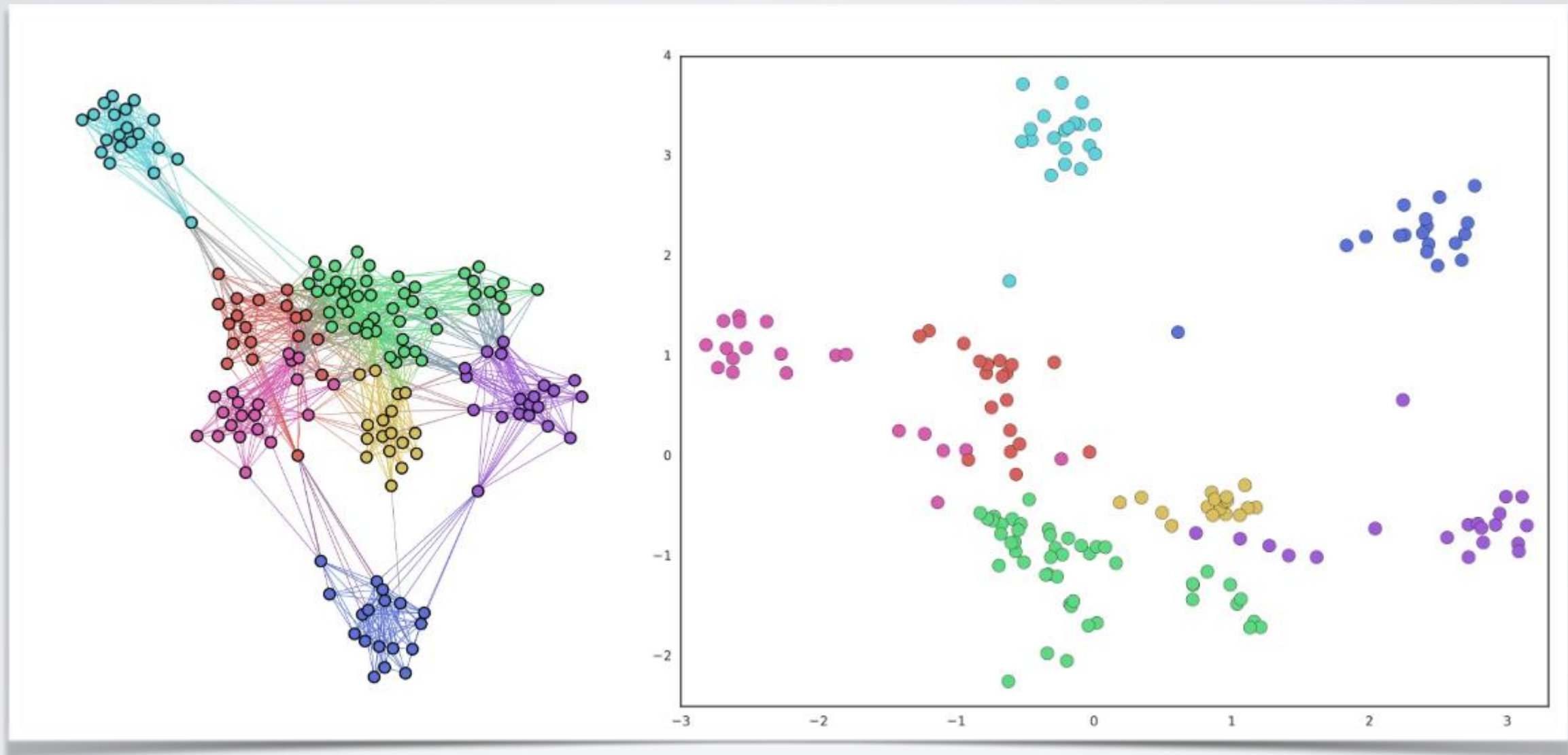
- Typical application case: Brain signal analysis
 - Distance is computed as signal correlation on fMRI, i.e., regional brain activity
 - => Time series to graph



ITEM-ITEM GRAPH

- We can use graphs as an alternative to dimensionality reduction for visualization
 - PCA / tSNE: project items in 2D, close items are similar
 - Some impossibilities, e.g., multiple semantics for words (“palm”: part of the hand, tree)
 - Networks can also be viewed in 2D and preserve the similarity information
- Approach:
 - 1) Compute the distance between elements
 - Euclidean
 - Cosine
 - 2) Keep as an edge values above a threshold

ITEM-ITEM GRAPH



Comparison PCA-graph representation

FEATURE-FEATURE GRAPH

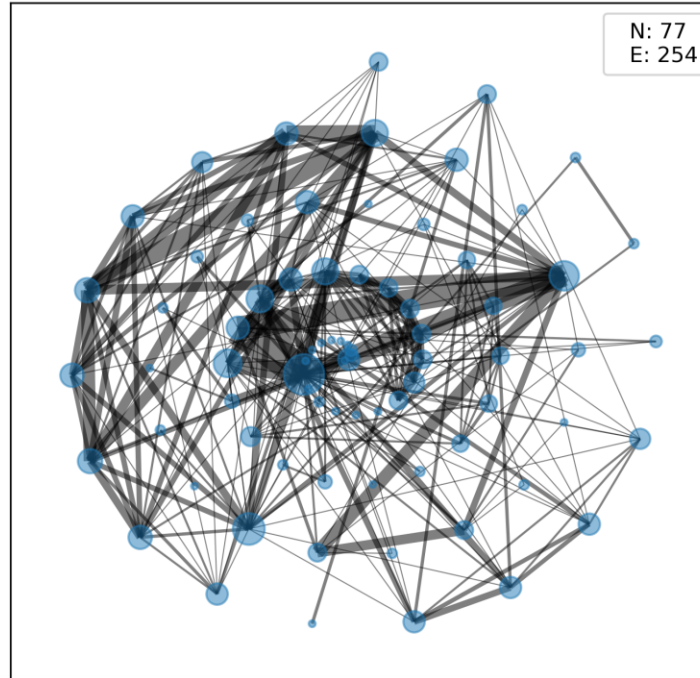
- Imagine an apartment dataset with variables surface, # rooms, etc.
 - Item-item: apartment as nodes, links represent similar apartments
 - Feature-feature: each feature is a node, edges represent relations/correlation
- Useful in particular when many variables
 - Recommendation
 - Biological data
 - etc.

BACKBONE EXTRACTION

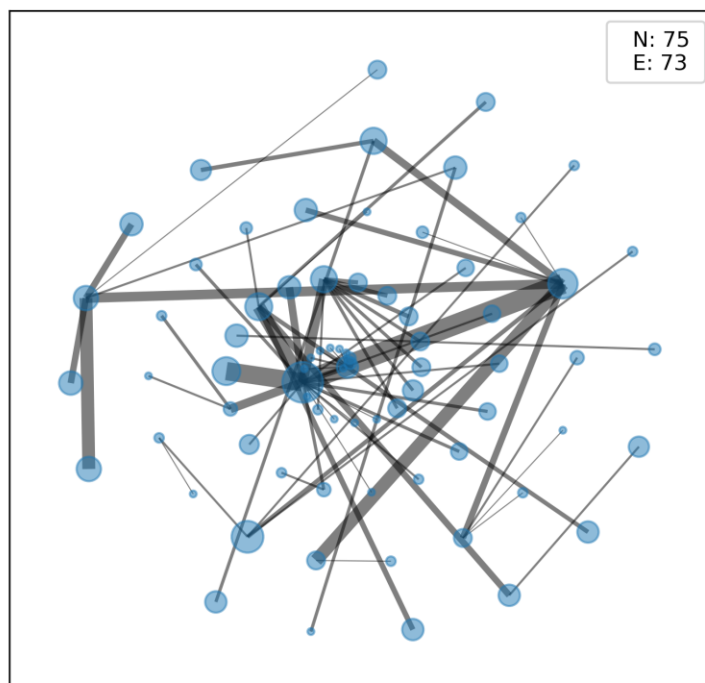
- In some cases, the network created might be too dense to be analyzed properly
 - Too low threshold: everything is connected
 - Too high: disconnected graph, most elements removed
- A solution is to use Backbone extraction
 - Methods that retain only the most important edges, based on different principles
 - e.g., <https://pypi.org/project/netbone/>

BACKBONE EXTRACTION

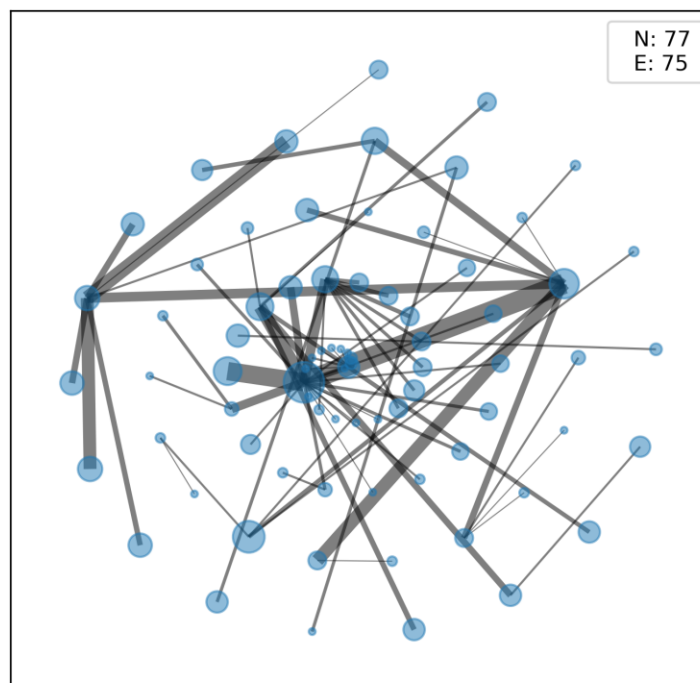
Les Misérables Original Network



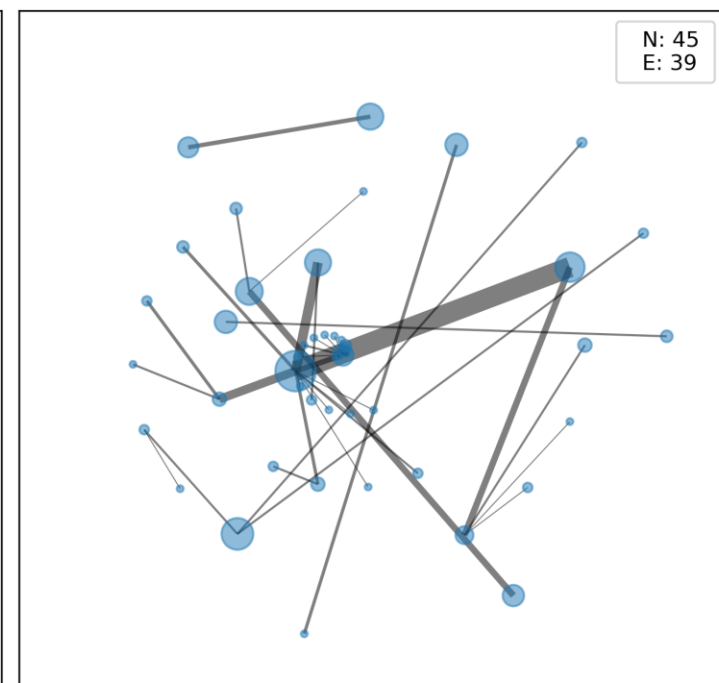
Boolean Filter



Threshold Filter



Fraction Filter

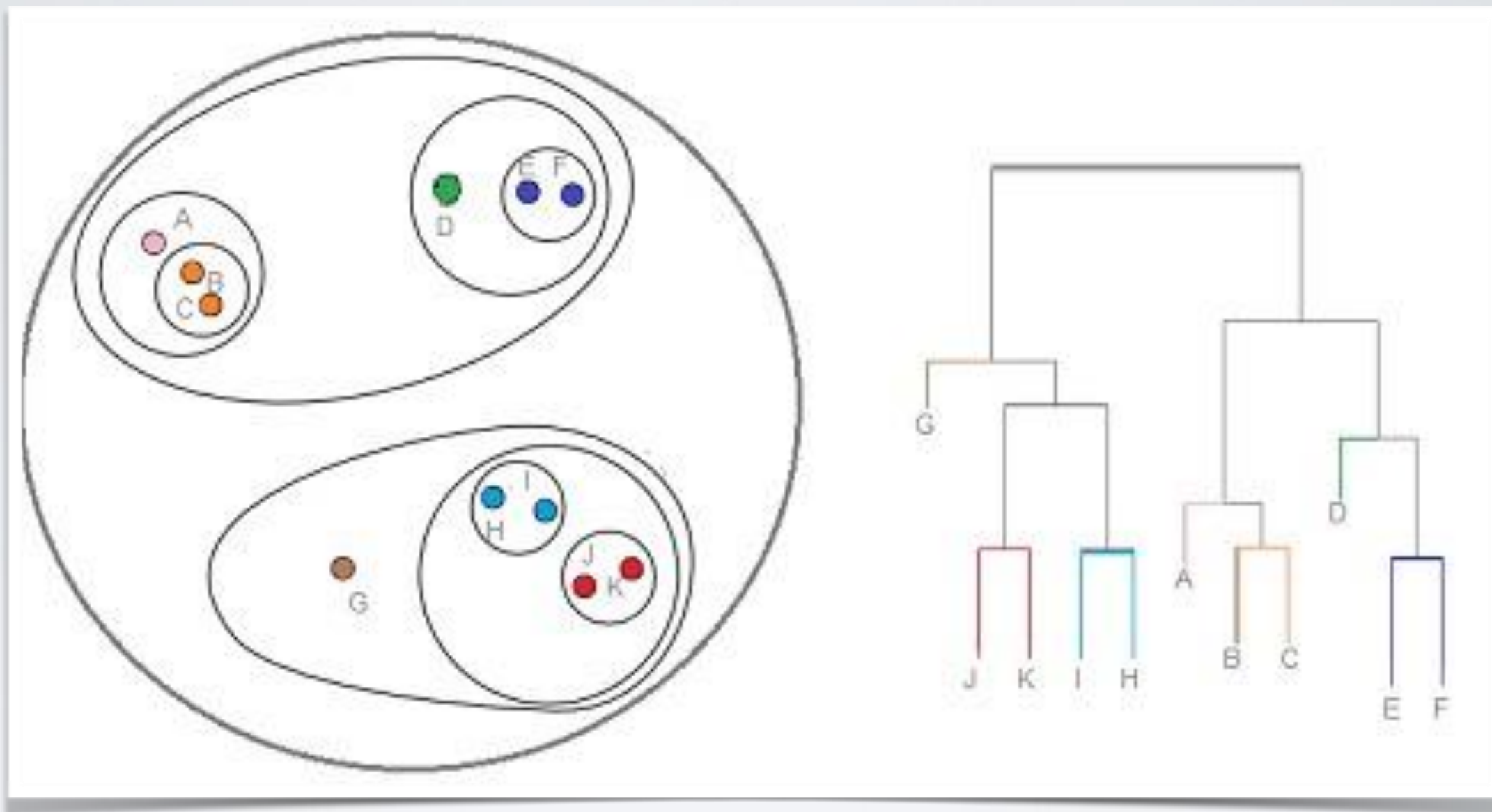


SPATIAL DATA ANALYSIS

AGGLOMERATIVE CLUSTERING

- Agglomerative clustering is (yet another) clustering method
- Define a notion of distance between two sets of points, e.g.
 - Minimal distance between sets elements
 - Average distance between elements
 - ...
- Start with each item in its own cluster
- **While** nb_cluster > 1
 - Merge the two closest cluster

DENDROGRAM



<https://www.statisticshowto.com/hierarchical-clustering/>

CLUSTER DISTANCES

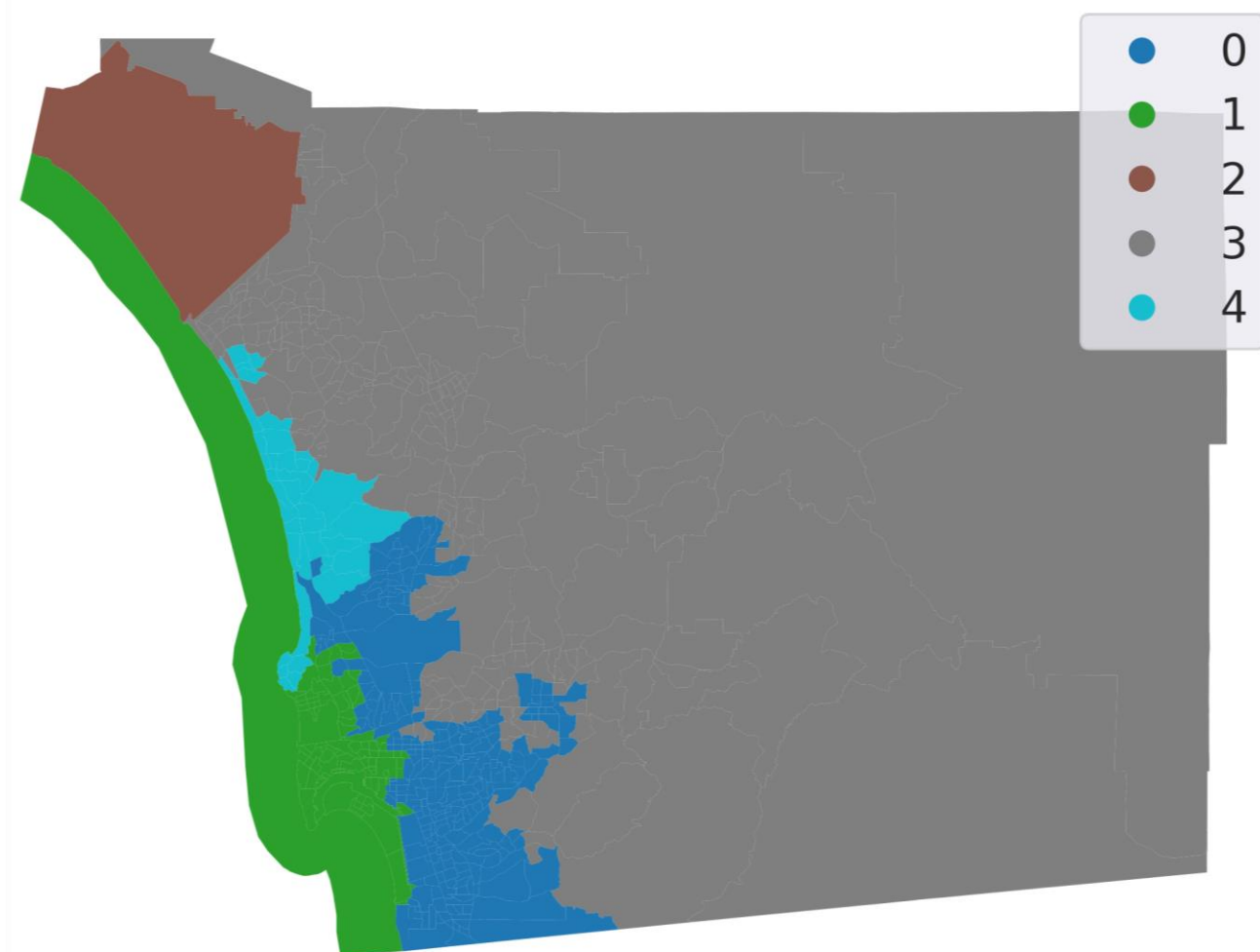
- Choose a distance function
 - Euclidean distance
 - Cosine distance
 - ...
- Choose a cluster distance strategy
 - **single** uses the minimum of the distances between all observations of the two sets.
 - **complete** or 'maximum' linkage uses the maximum distances between all observations of the two sets.
 - **average** uses the average of the distances of each observation of the two sets.
 - **ward** minimizes the variance of the clusters being merged. (Within-Cluster Sum of Squares)
 - $\Delta WCSS = WCSS_{new} - (WCSSC_1 + WCSSC_2)$
 - Similar objective than k-means, but more greedy

REGIONALIZATION

- Clustering: finding groups of similar observations
- If the data has a spatial structure, we might want the clusters to be contiguous in space
- => Add a spatial constraint

REGIONALIZATION

e.g., vote, weather...



REGIONALIZATION

- To discover spatial clusters, we want to allow merging only **spatially contiguous** clusters
- Solution: Connectivity matrix
 - A **graph** describing what element is a **neighbor** of another element.
 - Can merge only clusters with at least one edge between clusters

REGIONALIZATION



REGIONALIZATION

- Connectivity matrix (Binary graph)
 - Contiguity:
 - Contact between surface
 - Distance < threshold
 - KNN (K-nearest-neighbors)
- Spatial Weights Matrix (Weighted graph)
 - Put weights on edges
 - Inverse of the distance
 - Inverse of the squared distance...
 - Row normalized: sum of weights of neighbors=1

REGIONALIZATION

- Other methods
 - K-means with constraints
 - Multiple variants
 - DBSCAN: principle of a graph with threshold...

SPATIAL AUTOCORRELATION

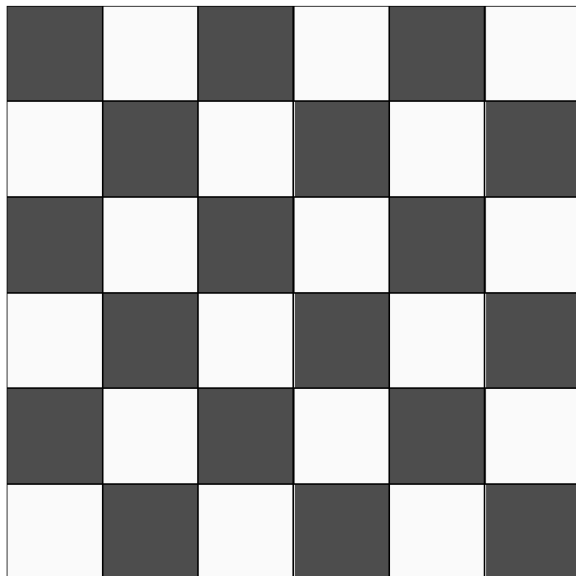
Global

INTUITION

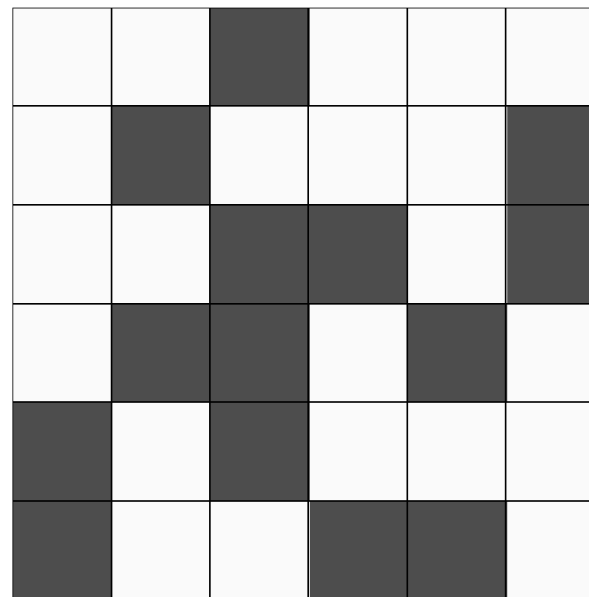
- Suppose you have attributes on observations
 - Binary (vote FOR/AGAINST, has covid cases or not, etc.)
 - Multi-label (candidate, type of apartments, etc.)
- Are those points distributed randomly/independently?
 - Or is there a correlation between the position of a point and the ones close to it
- Correlation between a variable and itself in space
 - => Spatial autocorrelation

INTUITION

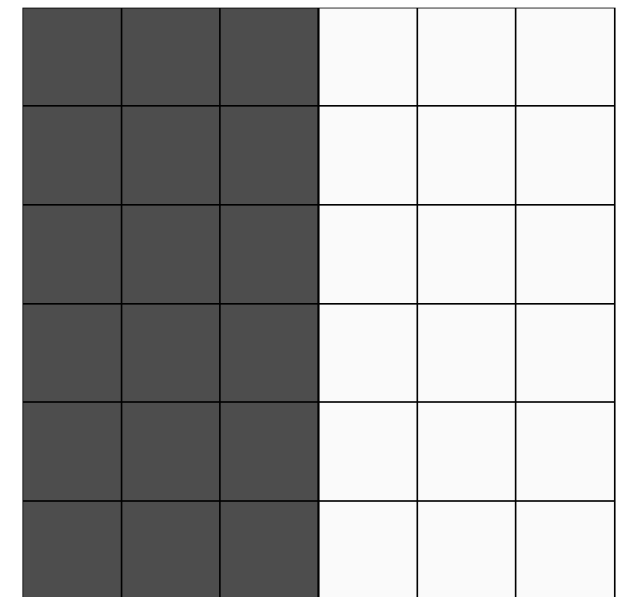
Negative spatial autocorrelation



No spatial autocorrelation



Positive spatial autocorrelation

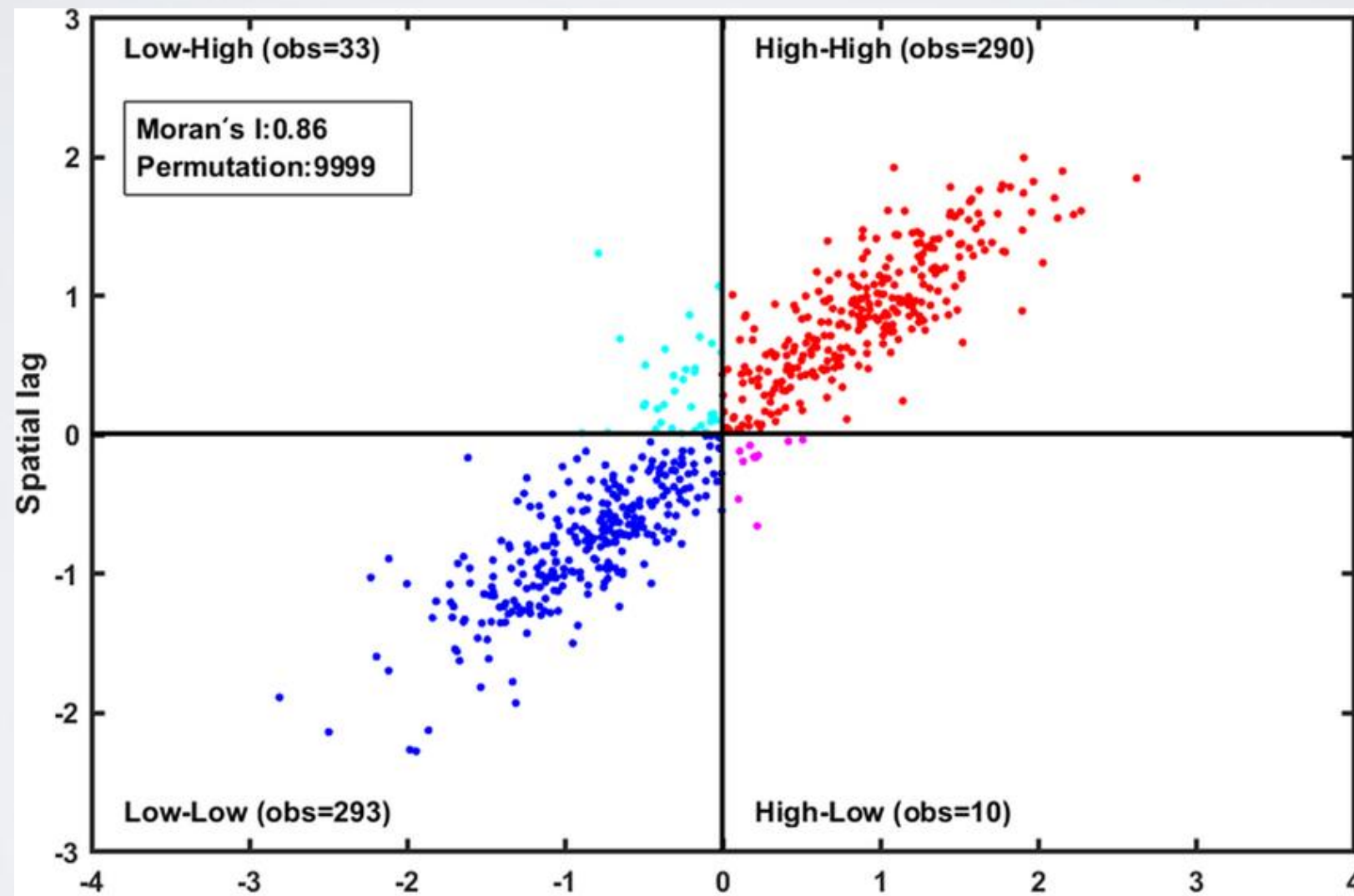


INTUITION

- Using a Spatial Weights Matrix
 - w_{ij} : weight of edge (i, j)
- Spatial lag: $y_i^{sl} = \sum_j w_{ij} y_j$
 - With y_j the variable of interest
- Weighted average of neighbors

MORAN'S PLOT

Plot relation between standardized values



Moran's I is the slope of a linear regression on this plot

LINEAR SPATIAL AUTOCORRELATION

- Compute Pearson's linear correlation between
 - Value for observation x
 - Spatial lag for observation x
- In practice, people rather use Moran's I
 - Generalization to take into account:
 - Different # of neighbors
 - Different weights
 - Slope of linear regression on Moran's plot

MORAN'S I

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij} z_i z_j}{\sum_i z_i^2}$$

- w_{ij} : weight of edge (i, j)
- z_i : value at i , standardized
- n : nb. of observations

SPATIAL AUTOCORRELATION

Local

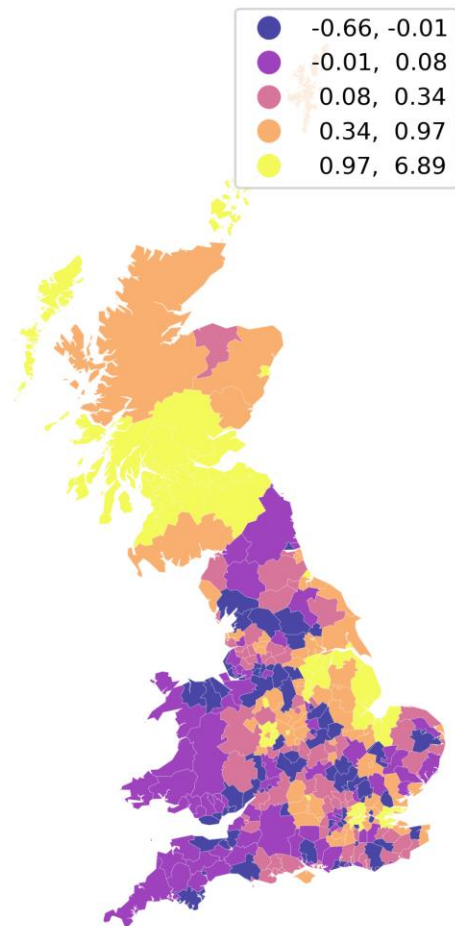
INTUITION

- Single scores are often misleading
- We can look at the details:
 - Where are positive/negative autocorrelations?
 - Where is the autocorrelation significant?
- Introduce LISA
 - Local Indicators of Spatial Association

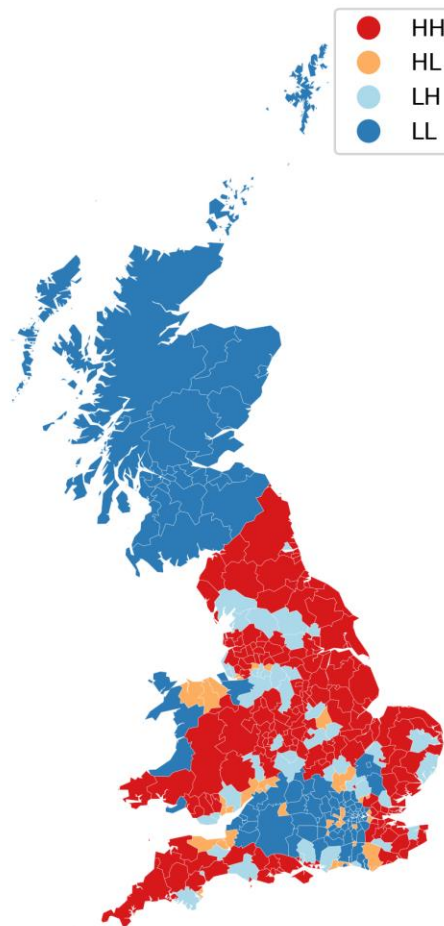
LISA

- 1) Compute significance: Moran's I_i
 - $I_i = \frac{z_i}{m_2} \sum_j w_{ij} z_j; m_2 = \frac{\sum_i z_i^2}{n}$
 - m_2 : variance of the variable of interest
 - z_i : standardized value
 - Positive value: positive spatial correlation at this point
 - Negative value: negative spatial correlation at this point
 - 0 or close to 0: no significant spatial autocorrelation
- Threshold on this value to decide significance

Brexit vote example (Support for Brexit)



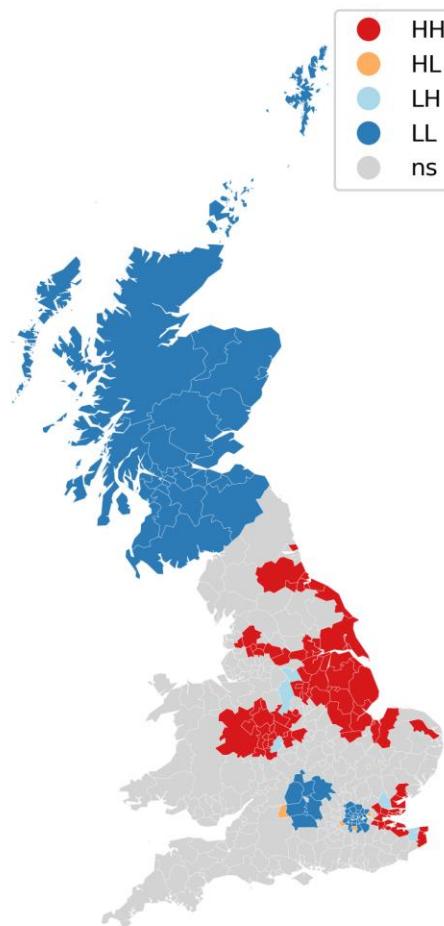
Local Statistics



Scatterplot Quadrant



Statistical Significance



Moran Cluster Map

HH: Hot spots
LL: Cold spots
LH: doughnuts
HL: diamonds in the rough

https://geographicdata.science/book/notebooks/07_local_autocorrelation.html